

Exercise (1). $x(t) = 10t^2$ / [2s-3s].

1) Average velocity. $v_{avg} = \frac{\Delta x}{\Delta t} = \frac{90-40}{3-2} \Rightarrow \boxed{v_{avg} = 50 \text{ m/s}}$

2) Average acceleration.

1st methode. $a = \frac{d^2x}{dt^2} = \frac{d^2}{dt^2}(10t^2) \Rightarrow \boxed{a = 20 \text{ m/s}^2}$

2nd methode. $a = \frac{\Delta v}{\Delta t} = \frac{50-40}{2,5-2} \Rightarrow \boxed{a = 20 \text{ m/s}^2}$

$\hookrightarrow \boxed{v = 20t}$

Exercise (2).

1) average velocity. $v_{avg} = \frac{\Delta x}{\Delta t} = \frac{2-8}{4-1,5} \Rightarrow \boxed{v_{avg} = -2,4 \text{ m/s}}$

2) instantaneous velocity. $v_{inst} = \frac{dx}{dt} = \frac{0-13,1}{3,5-0} \Rightarrow \boxed{v_{inst} = -3,7 \text{ m/s}}$

3) from the graph; the velocity will be zero ($v_{inst} = 0 \text{ m/s}$) when the slope is zero. this occurs for the point on graph where (x) has its minimum value; this is at $t = 4 \text{ s}$

Exercise (3).

1) velocity graph.

$t_0 = 0 \text{ (s)} \longrightarrow \boxed{v_0 = 2 \text{ m/s}}$

$t_1 = 8 \text{ (s)} \longrightarrow v_1 = a(t_1 - t_0) + v_0 = -0,25(8-0) + 2 \Rightarrow \boxed{v_1 = 0 \text{ m/s}}$

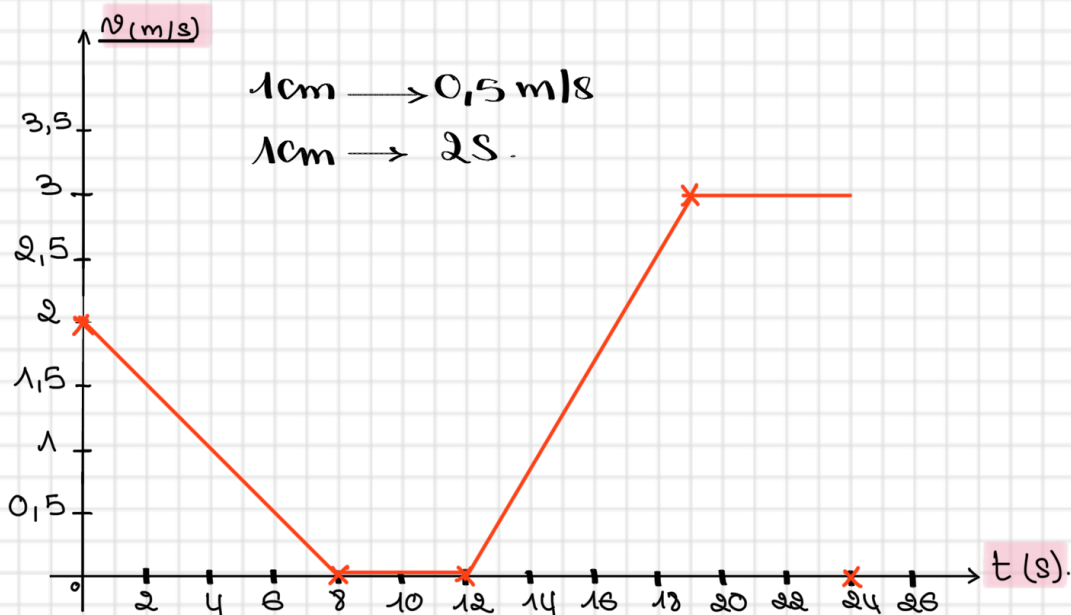
$t_2 = 12 \text{ (s)} \longrightarrow v_2 = a(t_2 - t_1) + v_0 \Rightarrow \boxed{v_2 = 0 \text{ m/s}}$ because $\boxed{a = 0 \text{ m/s}^2}$

$t_3 = 18 \text{ (s)} \longrightarrow v_3 = a(t_3 - t_2) + v_0 = 0,5(18-12) + 0 \Rightarrow \boxed{v_3 = 3 \text{ m/s}}$

$t_4 = 24 \text{ (s)} \longrightarrow v_4 = a(t_4 - t_3) + v_0 \Rightarrow \boxed{v_4 = 0 \text{ m/s}}$

$\forall t: 18 \rightarrow 24 \Rightarrow \boxed{v = 3 \text{ m/s}}$ because $\boxed{a = 0 \text{ m/s}^2}$

$\Rightarrow$ velocity graph
versus time



2) the position at $(t=8s)$: motion is non-uniform

$$X(t=8s) = \frac{1}{2} a(t=8s) t^2 + \cancel{v_0} \cdot t + \cancel{x_0} = \frac{1}{2} (-0,25) (8)^2 + 2 \times 8$$
$$\Rightarrow \boxed{X(t=8s) = 8 \text{ m}}$$

3) velocity of B.

* motion is uniform because $(v = \text{cte}) \Rightarrow X_B = v_B \cdot t + x_0 \Rightarrow 8 = v_B \times 8$
 $\Rightarrow \boxed{v_B = 1 \text{ m/s at } t = 8 \text{ s}}$

4) (B) will meet (A) between: 18,5 s $\in$ (18 s and 24 s)

* at $t = 8 \text{ s} \Rightarrow X_B = v_B \cdot t = 1 \times 18 \Rightarrow \boxed{X_B = 18 \text{ m}}$

$$X_A = \frac{1}{2} a(t_2 - t_1)^2 + v_0 (t_2 - t_1) + x_0 = \frac{1}{2} (0,5) (18 - 12)^2 + 0(18 - 12) + 8$$
$$\Rightarrow \boxed{X_A = 17 \text{ m}}$$

* $X_A = X_B$ ($t = 18 \rightarrow 24 \text{ s}$)

$$\Rightarrow v_A(t - t_0) + x_0 = v_0 \cdot t \Rightarrow v_A(t - 18) + 17 = v_0 \cdot t \Rightarrow 3(t - 18) + 17 = t$$
$$\Rightarrow 2t = 37 \Rightarrow \boxed{t = \frac{37}{2} = 18,5 \text{ (s)}}$$

5) Mobile (A) is the first because: $X_A = 23 \text{ m} > X_B = 20 \text{ m}$

* $X_A = v_A(t_2 - t_1) + x_0 = 3(20 - 18) + 17 \Rightarrow \boxed{X_A = 23 \text{ m}}$

* $X_B = v_B \cdot t = 20 \times 1 \Rightarrow \boxed{X_B = 20 \text{ m}}$

uniform motion

Exercise (4)

1) trajectory equation:

* $x(t) = 18t$ (1)

* $y(t) = -4,9t^2 + 4t$ (2)

- from (1) we find $\Rightarrow \boxed{t = \frac{x}{18}}$

- By compensation in (2) we get:

$$\Rightarrow y(t) = -4,9 \left(\frac{x}{18} \right)^2 + 4 \left(\frac{x}{18} \right) \Rightarrow \boxed{y(t) = -0,015x^2 + 0,222x}$$

* type of motion is projectile

2) position vector:

* $\vec{r}(t) = x\vec{i} + y\vec{j} \Rightarrow \boxed{\vec{r}(t) = 18t\vec{i} + (-4,9t^2 + 4t)\vec{j}}$

3) velocity vector :

$$\vec{v}(t) = \frac{d\vec{r}(t)}{dt} \Rightarrow \boxed{\vec{v}(t) = 18\vec{i} + (4 - 9,8t)\vec{j}}$$

4) acceleration vector :

$$\vec{a}(t) = \frac{d\vec{v}(t)}{dt} \Rightarrow \boxed{\vec{a}(t) = -9,8\vec{j}}$$

5) calculate at t = 3 s :

$$\vec{r}(t=3s) = (18 \times 3)\vec{i} + (4 \times 3 - 4,9 \times 3^2)\vec{j} = 54\vec{i} + (12 - 44,1)\vec{j} \\ \Rightarrow \boxed{\vec{r}(t=3s) = 54\vec{i} - 32,1\vec{j}}$$

$$\vec{v}(t=3s) = 18\vec{i} + (4 - 9,8 \times 3)\vec{j} \Rightarrow \boxed{\vec{v}(t=3s) = 18\vec{i} - 25,4\vec{j}}$$

$$\vec{a}(t=3s) = -9,8\vec{j}$$

Exercise(5) : $v_0 = 200 \text{ m/s}$; $\theta = 40^\circ$

* we have projectile motion.

1) velocity :

$$\vec{v}(t) = \frac{dx}{dt}\vec{i} + \frac{dy}{dt}\vec{j} = v_0 \cos \theta \vec{i} + (v_0 \sin \theta - gt)\vec{j} \Rightarrow \boxed{\vec{v}(t) = 153,2\vec{i} - 67,44\vec{j}}$$

$$\Rightarrow v = \sqrt{(153,2)^2 + (-67,44)^2} \Rightarrow \boxed{v = 167,4 \text{ m/s}}$$

2) position :

$$\vec{r}(t) = x\vec{i} + y\vec{j} = (v_0 \cos \theta \cdot t)\vec{i} + \left(v_0 \sin \theta \cdot t - \frac{1}{2}gt^2\right)\vec{j} \\ \Rightarrow \boxed{\vec{r}(t) = 3064\vec{i} + 611,2\vec{j}}$$

3) Range : By using equation of path :

$$t = \frac{x}{v_0 \cos \theta} \Rightarrow y = x \tan \theta - \frac{gx^2}{2v_0^2 \cos^2 \theta} = 0 \Rightarrow \boxed{x = 4020,6 \text{ m}}$$

4) max height : we have $v_y = 0 \text{ m/s}$

$$H = \frac{v_0^2 \sin^2 \theta}{2g} \Rightarrow \boxed{H = 843,22 \text{ m}}$$

5) time of flight :

$$t_f = \frac{2v_0 \sin \theta}{g} \Rightarrow \boxed{t_f = 26,24 \text{ (s)}}$$

Exercise(6) : $\theta(t) = 3t^2 + 2t$

1) Angular velocity

$$\vec{\omega}(t) = \dot{\theta}(t) = \frac{d\theta(t)}{dt} \Rightarrow \boxed{\omega(t) = 6t + 2}$$

$$\text{at } (t=4s) \Rightarrow \omega(t=4s) = 6 \times 4 + 2 = \boxed{26 \text{ rad/s}}$$

2) Angular acceleration

$$\alpha(t) = \dot{\omega}(t) = \frac{d^2\theta(t)}{dt^2} = \frac{d\omega(t)}{dt} \Rightarrow \boxed{\alpha(t) = 6 \text{ rad/s}^2} \rightarrow \alpha = \text{cte}$$

Exercise (7)

$$\alpha(t) = 12t^2 - 48t + 16$$

1) Angular velocity

$$\omega = \int \alpha(t) \cdot dt = \int (12t^2 - 48t + 16) dt = 4t^3 - 24t^2 + 16t + C$$

$$\text{at } (t=0s) \text{ we have } \omega(t=0) = 0 \text{ rad/s then } \boxed{C=0}$$

$$\text{So } \boxed{\omega(t) = 4t^3 - 24t^2 + 16t}$$

2) Angular position

$$\theta(t) = \int \omega(t) \cdot dt = \int (4t^3 - 24t^2 + 16t) dt = t^4 - 8t^3 + 8t^2 + C'$$

$$\text{at } (t=0s) \text{ we have } \theta(t=0) = 0 \text{ rad then } \boxed{C'=0}$$

$$\text{So } \boxed{\theta(t) = t^4 - 8t^3 + 8t^2}$$

3) components of acceleration (tangential and centripetal)

tangential (a_t)

$$a_t = r \cdot \alpha(t) = 1,3(12t^2 - 48t + 16)$$

$$\Rightarrow \boxed{a_t = 15,6t^2 - 62,4t + 20,8}$$

centripetal (a_c)

$$a_c = \frac{v^2}{r} = \frac{(r\omega)^2}{r} = r\omega^2$$

$$\Rightarrow \boxed{a_c = 1,3 \times (4t^3 - 24t^2 + 16t)^2}$$

Exercise (8)

$$x(t) = -5 \sin \omega t \quad | \quad y(t) = 4 - 5 \cos \omega t$$

1) trajectory equation

$$x^2 = 25 \sin^2 \omega t \quad (1) \quad | \quad (y-4)^2 = 25 \cos^2 \omega t \quad (2)$$

by adding (1) and (2) we get

$$x^2 + (y-4)^2 = 25 \Rightarrow \boxed{(x-0)^2 + (y-4)^2 = 5^2}$$

$\Rightarrow$ it is uniform circle motion with radius ($R=5m$) centered at $(0,4)$.

2) position vector

$$\vec{r}(t) = x\vec{i} + y\vec{j} \Rightarrow \boxed{\vec{r}(t) = (-5 \sin \omega t)\vec{i} + (4 - 5 \cos \omega t)\vec{j}}$$

* velocity vector

$$\vec{v}(t) = \frac{d\vec{r}(t)}{dt} = \frac{d}{dt}((-5\sin\omega t)\vec{i} + (4-5\cos\omega t)\vec{j}) \Rightarrow \vec{v}(t) = (-5\omega\cos\omega t)\vec{i} + (5\omega\sin\omega t)\vec{j}$$

* Acceleration vector

$$\vec{a}(t) = \frac{d\vec{v}(t)}{dt} = \frac{d}{dt}((-5\omega\cos\omega t)\vec{i} + (5\omega\sin\omega t)\vec{j}) \Rightarrow \vec{a}(t) = (5\omega^2\sin\omega t)\vec{i} + (5\omega^2\cos\omega t)\vec{j}$$

3) the expressions of velocity and acceleration vectors at any time $t > 0$.

I. in polar system $\rightarrow \vec{u}_0$ and $\vec{u}_r$

we have $x = -5\sin\omega t$ and $y = 4 - 5\sin\omega t$

* the position of point M is $(R, \theta) \Rightarrow M(5, \frac{\pi}{2} - \omega t)$

$$R = 5\text{m} \quad | \quad \theta = \tan^{-1}\left(\frac{y-4}{x}\right) = \frac{\pi}{2} - \omega t$$

* velocity is $\vec{v} = \frac{d(r\vec{u}_r)}{dt} = r\frac{d\vec{u}_r}{dt} + \vec{u}_r\frac{dr}{dt} = [r\omega\vec{u}_0] = [5\omega\vec{u}_0]$

* acceleration is $\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}(r\omega\vec{u}_0) = \frac{dr}{dt}(\omega\vec{u}_0) + \frac{d\omega}{dt}(r\vec{u}_0) + \frac{d\vec{u}_0}{dt}(r\omega)$
 $= [\underbrace{r\alpha\vec{u}_0}_{a_t} - \underbrace{r\omega^2\vec{u}_r}_{a_r = a_n}]$ with $\alpha = \text{Angular velocity}$

II. intrinsic coordinates $\rightarrow \vec{u}_t$ and $\vec{u}_n$

* velocity is $\vec{v} = v\vec{u}_t = r\omega\vec{u}_t = [5\omega\vec{u}_t]$

* acceleration is $\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}(r\omega\vec{u}_t) = \frac{dr}{dt}(\omega\vec{u}_t) + \frac{d\omega}{dt}(r\vec{u}_t) + r\omega\frac{d\vec{u}_t}{dt}$

$$= [\underbrace{r\alpha\vec{u}_t}_{a_t} + \underbrace{r\omega^2\vec{u}_n}_{a_n = \text{normal}}]$$

Exercise(9) = (1/100) scale

$$\omega(t) = \begin{cases} 2t; & t \leq 2 \\ \pi; & t \geq 2 \end{cases}$$

1) the value and unit of α and it's physical meaning

we have $\omega(t) = 2t \Rightarrow \alpha = \frac{d\omega(t)}{dt} = \frac{2}{1} \frac{\text{rad/s}}{\text{s}} \Rightarrow \alpha = \text{Angular acceleration}$
 $\uparrow$ angular velocity $\uparrow$ rad/s²

so the angular velocity ω it's value is at $t = 2 \Rightarrow 2t = \pi \Rightarrow 2\alpha = \pi \Rightarrow \alpha = \frac{\pi}{2}$

and its unit is rad/s^2

* the radius (R) is $R = 1,4 \times 100 = 140\text{cm} = 1,4\text{m}$

* the angular position is $\theta = \int \omega(t) dt = \begin{cases} \int 2t dt = \frac{\pi}{2} t dt; & t \leq 2\text{s} \\ \int \pi dt; & t \geq 2\text{s} \end{cases}$

$$\Rightarrow \theta = \begin{cases} \frac{\pi}{4} t^2 + C ; t \leq 2s \\ \pi \cdot t + C' ; t \geq 2s \end{cases} \quad \left| \begin{array}{l} \text{at } (t=0) \Rightarrow \theta_0 = 0 \text{ rad so } \boxed{C=0} \\ \text{at } (t=2s) \Rightarrow \frac{\pi}{4} t^2 = \pi t + C' \Rightarrow \pi = 2\pi + C' \text{ so } \boxed{C' = -\pi} \end{array} \right.$$

then we get:

$$\theta = \begin{cases} \frac{\pi}{4} t^2 ; t \leq 2s \\ \pi t - \pi ; t \geq 2s \end{cases}$$

2) the curved abscissa (coordinates).

$$\widehat{M_0 M} = r \theta (t=2.5s) = 140 \times \frac{3\pi}{2} = \boxed{210\pi \text{ cm} = 2.1\pi \text{ m}}$$

$$r = 140 \text{ cm} \quad \theta = \pi \times 2.5 - \pi = \frac{3\pi}{2} \text{ rad.}$$

3) components of velocity vector:

we have: $\vec{v} = v_r \cdot \vec{u}_r + v_\theta \cdot \vec{u}_\theta$

$$v_r = \frac{dr}{dt} = \boxed{0 \text{ m/s}} \text{ because } r = \text{cte.}$$

$$v_\theta = r \frac{d\theta}{dt} = r\omega = \begin{cases} r\alpha t = r \frac{\pi}{2} \cdot t, & t \leq 2s \\ \pi \cdot r, & t \geq 2s \end{cases}$$

4) intrinsic components (a_t) and (a_n):

$$\begin{array}{l} \ast a_t = \frac{dv_\theta}{dt} = r \frac{d\omega}{dt} = \boxed{r\alpha} \\ \ast a_n = \frac{v_\theta^2}{r} = \boxed{r \cdot \omega^2} \end{array} \quad \left| \begin{array}{l} \ast \text{for } (t \leq 2s) = a_t = r \cdot \alpha = \boxed{r \cdot \frac{\pi}{2}} \text{ (m/s}^2\text{)} \\ a_n = r \cdot \omega^2 = r \cdot \left(\frac{\pi}{2} \cdot t\right)^2 = \boxed{\frac{\pi}{4} r^2 t^2} \text{ (m/s}^2\text{)} \\ \ast \text{for } (t \geq 2s) = a_t = \boxed{0} \text{ (m/s}^2\text{)} \\ a_n = r \cdot \omega^2 = \boxed{r \cdot \pi^2} \text{ (m/s}^2\text{)}. \end{array} \right.$$

* transversal (a_θ) and radial (a_r):

$$\begin{array}{l} a_\theta = r \cdot \frac{d\omega}{dt} = r\alpha \quad \left| \quad \vec{a}_r = -r\omega^2 \cdot \vec{u}_r \right. \\ \ast \text{for } (t \leq 2s) = a_\theta = \boxed{r \cdot \frac{\pi}{2}} \text{ m/s}^2 \quad \left| \quad \ast \text{for } (t \geq 2s) = a_\theta = \boxed{0} \text{ m/s}^2 \right. \\ \vec{a}_r = \boxed{-\frac{\pi}{4} r^2 t^2 \cdot \vec{u}_r} \quad \left| \quad \vec{a}_r = \boxed{-r \pi^2 \cdot \vec{u}_r} \right. \end{array}$$

5) the equation of motion:

$$x_A = x_M + v_A(t-2s)$$

$\hookrightarrow x_M = r \cdot \cos \theta$; at $(t=2s)$ we have $\theta = \pi$ so $x_M = r \cdot \cos(\pi) \Rightarrow \boxed{x_M = -r}$ which means $\boxed{x_A = -r + v_A(t-2)}$

* velocity v_A after ($t = 2\text{ s}$) mobile (A) has the same velocity with mobile (M) so $\boxed{v_A = r \cdot \pi}$

* acceleration a we have =

$$a = \frac{dv_A}{dt} = \frac{d(r\pi)}{dt} = \boxed{0 \text{ m/s}^2}$$

* period (T) $\omega = \frac{2\pi}{T} \Rightarrow T = \frac{2\pi}{\omega} = \frac{2\pi}{\pi} \Rightarrow \boxed{T = 2\text{ s}}$